

Roll No.

TPE-002

**B. TECH. (ECE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

OPTICAL FIBER COMMUNICATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

1. (a) Draw and explain the general block diagram of optical fiber communication. (CO1)

(b) Define the relative refractive index difference for an optical fiber and show how it may be related to the numerical aperture. (CO1)

(c) A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.50 and a cladding refractive index of 1.47.

Determine : (CO1)

(i) The critical angle at the core-cladding interface

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- (ii) The Numerical Aperture of the fiber
 - (iii) The acceptance angle in air for the fiber
2. (a) Briefly describe the scattering losses that occur in optical fiber. (CO2)
- (b) Discuss the linear scattering losses in optical fibers with respect to : (CO2)
- (i) Mie Scattering
 - (ii) Stimulated Brillouin Scattering
- (c) The numerical input/output mean optical power ratio in a 1 km, length of optical fiber is found to be 2.5. Calculate the received mean optical power when a mean optical power of 1 MW is launched into a 5-km length of the fiber (assuming no joint or connectors). (CO2)
3. (a) What are the requirements of laser action ? Explain spontaneous and stimulated emission. (CO3)
- (b) What do you understand by optical modulator ? List the various modes of Laser diode. (CO3)
- (c) A forward biased LED has a voltage drop of 1.5 volts. If the battery voltage is 6 volts, calculate the resistance to be connected in series. if the current through the LED is 15 mA, how much power is dissipated in the resistor ? (CO3)
4. (a) Define photocarriers and photocurrent. What are the advantages of photodiodes ? (CO4)
- (b) Define avalanche effect. What are the drawbacks of Avalanche photodiode ? (CO4)

(3)

- (c) A photodiode has a quantum efficiency of 65% when photons of energy 1.5×10^{-19} J are incident upon it.

Calculate :

(CO4)

- (i) photodiode operating wavelength
- (ii) the incident optical power required to obtain a photocurrent of $2.5 \mu\text{A}$ when the photodiode is operating as described above.

5. (a) What is a preamplifier ? Give the classification of preamplifier.

(CO5 & CO6)

- (b) Write the characteristics as well as advantages EDFA. (CO5 & CO6)

- (c) A four port multimode fiber FBT (fused bioconical taper) coupler has $70 \mu\text{W}$ optical power launched into port one. The measured output powers at ports two, three and four are 0.004, 26 and $27.5 \mu\text{W}$ respectively. Determine : What are the coupling ratio, excess loss, insertion loss between the input and output ports. (CO5 & CO6)

Roll No.

TPE-004

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

DIGITAL SYSTEM USING VERILOG

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Create your own 2-input Verilog gates called my_or, my_and and my_not from 2-input nand gates using gate level modelling. Check the functionality of these gates with a stimulus module. (CO1, CO2)
- (b) Discuss the basic components of a module. Which components are mandatory and why ? (CO1, CO2)
- (c) What are compiler directives ? Explain any *two* compiler directives with suitable examples. (CO1, CO2)

P. T. O.

2. (a) In data flow modelling, give the Verilog description of JK flip-flop. Use the JK flip-flop instances to obtain D flip-flop. Write the test bench to verify the working of D flip-flop. (CO1, CO2)
- (b) Explain the following operators with the help of suitable examples : (CO1, CO2)
- (i) Concatenation
 - (ii) Replication
 - (iii) Conditional
 - (iv) Reduction
 - (v) Shift
- (c) What is the purpose of generate statement ? Give the three methods to create generate statement. Explain any *two* with example. (CO1, CO2)
3. (a) Differentiate between the following : (CO2, CO3)
- (i) Blocking and Non-Blocking assignment
 - (ii) Sequential and Parallel block
 - (iii) While loop and For loop
 - (iv) Forever and Repeat Statements
 - (v) Initial and always
- (b) Explain Task and Function in Verilog giving one example code for each. Highlight the difference between task and function. (CO2, CO3)
- (c) Write the Verilog description of Universal Shift Register and verify its working using suitable test bench. (CO2, CO3)
4. (a) What is UDP in Verilog ? How many types of UDP are there ? Explain the generic structure of UDP. (CO4)

(3)

- (b) Define a positive edge triggered D flip-flop with clear as a UDP. Signal clear is active low signal. Use shorthand notation wherever possible.

(CO4)

- (c) What is switch level modelling ? Explain with the help of a suitable example.

(CO4)

5. (a) What is a testbench ? Explain the basic structure of a testbench. Write testbench to verify working of exclusive or gate.

(CO5, CO6)

- (b) Write a test bench for the following :

(CO5, CO6)

(i) 4-Binary to Gray Code Converter

(ii) 8-bit BCD Adder Circuit

- (c) Write short note on Object Oriented Programming.

(CO5, CO6)

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Roll No.

TPE-007

**B. TECH. (ECE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022
SEMICONDUCTOR DEVICES AND MODELING**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Derive the expression for contact potential of p - n junction using energy band diagram. (CO1)
- (b) Explain the variation of electron concentration with temperature for n -type semiconductor. (CO1)
- (c) Explain in detail about formation of energy band. (CO1)
2. (a) Explain the effect of gate common terminal (V_{CB}) voltage of three terminal MOS structure on : (CO2)
 - (i) Flatband condition
 - (ii) Accumulation
 - (iii) Depletion and inversion

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(2)

- (b) Explain flatband voltage in detail along with expressions. Calculate the flatband voltage for a p -type body with $N_A = 10^{18} \text{ cm}^{-3}$, a SiO_2 insulator with a thickness $t_{ox} = 2 \text{ nm}$, and a n -type polysilicon gate with $N_D = 10^{20} \text{ cm}^{-3}$. The interface charge $Q_o = 10^{-8} \text{ C/cm}^{-2}$ and $\phi_{ms} = -1.036 \text{ V}$. (CO2)
- (c) Write short notes on the following : (CO2)
- (i) Body effect in MOS Transistor
 - (ii) Drain-Induced Barrier Lowering (DIBL)
3. (a) Explain SOI structures in detail. Mention the advantages of SOI structures. (CO3)
- (b) Discuss the hot carrier effect and impact ionization in a MOSFET. (CO3)
- (c) Describe the Fin-FET structures and explain its working and advantage in detail. (CO3)
4. (a) Write a SPICE code of enhancement mode MOSFET circuit for DC analysis. (CO3, CO4)
- (b) Explain the principle of circuit simulation and its objectives. (CO3, CO4)
- (c) Differentiate between partially depleted and fully depleted SOI MOSFETs. (CO3, CO4)
5. (a) Describe the working principle of hetero-junction bipolar transistors. (CO5, CO6)
- (b) Explain HEMTs in detail and differentiate between HBTs and HEMTs. (CO5, CO6)
- (c) Explain three-terminal MOS structure with proper band diagram. (CO5, CO6)

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TPE-013

**B. TECH. (ECE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

SATELLITE COMMUNICATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) The orbit of an earth orbiting satellite has an eccentricity of 0.15 and a semi-major axis of 9000 km. Given that, the radius of earth 6371 km and μ is $3.986 \times 10^{14} \text{ m}^3/\text{s}^2$, determine : (CO1)

(i) The Periodic Time

(ii) The Apogee Height

(iii) The Perigee Height

(b) A satellite is orbiting in geosynchronous orbit of radius 4150 km. Find the velocity and time of the orbit. What will be the change of velocity if the radius reduces to 36000 km ? If $g_0 = 398600.5 \text{ km}^2/\text{s}^2$. (CO1)

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- (c) Explain the Kepler's law of planetary motion. Briefly discuss the various types of orbital elements. (CO1)
2. (a) A satellite at a distance of 40,000 km from a point on earth's surface, radiates a power of 200 mW from an antenna with a gain of 17 dB in the direction of the observer. Find the flux density at the receiving point and the power received by an antenna with an effective area of 10 m^2 . (CO2)
- (b) Explain the Communication Subsystem. Briefly discuss the various types of Transponders. (CO2)
- (c) Describe the telemetry, tracking and command (TTC&M) subsystems of a satellite system. (CO2)
3. (a) A radio link uses a pair of 2 m dish antenna with an efficiency of 60% each, as transmitting and receiving antenna. Other specifications of the link are; transmitted power is 1 dBW, Carrier frequency is 4 GHz, Distance of receiver from the transmitter is 150 m. Calculate : (CO3)
- (i) Free Space Loss
 - (ii) Power gain of each antenna
 - (iii) Received power in dBW
- (b) Write short notes on the following : (CO3)
- (i) Explain the effect of rain on satellite link reliability.
 - (ii) Input/Output Back-off
- (c) A satellite stationed at a distance 38000 km from the surface of the earth radiates a power of 3 watts, in the direction of an earth station. Assuming the satellite antenna gain to be 20 dB. If the operating frequency is 10 GHz and gain of receiving antenna is 50 dB calculate the power received. (CO3)

(3)

4. (a) Explain the working of TWT amplifier in detail with the help of a diagram. (CO4)
- (b) Explain different multiple access techniques in detail. (CO4)
- (c) If a satellite has a transponder having 50 MHz BW and 1023 barker code is used for CDMA by various earth stations, the chip rate is 32 MCPS. Determine the number of accesses for an SNR of 12 dB. Also, find the total occupancy of transponder if BPSK is used. (CO4)
5. (a) What is the difference between regional and global navigation systems ? Explain the working principle of satellite navigation. (CO5, CO6)
- (b) What do you understand by the DTH System ? Explain the working of DTH. (CO5, CO6)
- (c) Explain the VSAT System in detail with its applications. (CO5, CO6)